



Maritime &
Coastguard
Agency

Guidance

MIN 715 (M) Carriage of Cargoes - Adoption of Amendment 07-23 of the International Maritime Solid Bulk Cargoes (IMSBC) Code and Amendment 42-24 of the International Maritime Dangerous Goods (IMDG) Code

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Summary

The purpose of this Marine Information Note (MIN) is to advise of an amendment (07-23) to the International Maritime Solid Bulk Cargoes (IMSBC) Code and its application date. This amendment makes a number of changes to the IMSBC Code, some of which are outlined below, and also adds new products which have not previously been included in Appendix 1 (Individual Schedules of Solid Bulk Cargoes) of the Code.

This MIN also advises of an amendment (42-24) to the International Maritime Dangerous Goods (IMDG) Code and its application dates. This amendment makes a number of changes to the IMDG Code, of which some of the key amendments are outlined below.

1. Introduction

The International Maritime Solid Bulk Cargoes (IMSBC) Code

1.1. The IMSBC Code became mandatory on an international basis when it entered into force on 1 January 2011. The Code facilitates the safe loading, stowage and shipment of solid bulk cargoes by providing information on the dangers associated with the shipment of certain types of solid bulk cargoes and instructions on the procedures to be adopted when the shipment of a solid bulk cargo is considered. The IMSBC Code is implemented in the UK by the Merchant Shipping (Carriage of Cargoes) Regulations 2024 (SI No.2024/637) and the Merchant Shipping (Carriage of Dangerous Goods and Harmful Substances) (Amendment) Regulations 2024 (SI No. 2024/636).

1.2. In order to keep the IMSBC Code up to date, the Code is updated in the International Maritime Organization (IMO) every two years. The purpose of this MIN is to advise of the latest amendments to the IMSBC Code that have been adopted in the IMO by means of Resolution MSC.539(107) specifying forthcoming amendments (07-23) to the IMSBC Code.

1.3. The amendments (07-23) to IMSBC Code will become mandatory on an international basis from 1 January 2025, and mandatory in the UK on the same date. A summary of these amendments is set out in Section 2 below.

The International Maritime Dangerous Goods (IMDG) Code

1.4. The IMDG Code provides the international standard for the carriage of dangerous goods and marine pollutants by sea. The Code is made mandatory on an international basis by Chapter VII of the International Convention for the Safety of Life at Sea (SOLAS) and also by Annex III of the International Convention for the Prevention of Pollution from Ships (MARPOL). The IMDG Code provides the international regulatory framework for the carriage of dangerous goods and marine pollutants by sea. It makes mandatory provision for all aspects of the transport by sea of dangerous goods, including the classification, notification, packaging, labelling, stowage and segregation of dangerous goods to enable their safe transport. The IMDG Code is implemented in the UK by the Merchant Shipping (Carriage of Dangerous Goods and Harmful Substances) (Amendment) Regulations 2024 (SI No.2024/636).

1.5. In order to keep the IMDG Code up to date and harmonised with the requirements for the carriage of dangerous goods in other modes of transport, the Code is updated in the IMO every two years. The purpose of this MIN is to advise of amendments to the IMDG Code that have been

adopted in the IMO by means of Resolution MSC.556(108), specifying forthcoming amendment 42-24 to the IMDG Code.

1.6. Amendment 42-24 can be used on a voluntary early adoption basis from 1 January 2025 and becomes mandatory internationally and in the UK on 1 January 2026. Consequential amendments to the Revised Emergency Response Procedures for Ships Carrying Dangerous Goods (EMS guide) (MSC.1/Circ.1588/Rev.2) were also approved in the IMO.

1.7. A summary of some of the key changes to the IMDG Code in amendment 42-24 are set out in Section 3 below.

Ambulatory Reference Provisions

1.8. Ambulatory Reference (AR) provision is a tool used in some UK secondary legislation which enables the automatic implementation of amendments to international instruments into UK legislation. Where the UK legislation refers to an international instrument, such as a provision in a convention or a code forming part of that convention (for example, the IMSBC Code or the IMDG Code), this reference is 'ambulatory', that is, a reference to the most up to date version of that provision or code.

1.9. AR provisions help to ensure that UK legislation does not fall behind international standards. Amendment 07-23 to the IMSBC Code will become mandatory in the UK by way of the ambulatory reference provision in the Merchant Shipping (Carriage of Cargoes) Regulations 2024, and amendment 42-24 to the IMDG Code will become mandatory in the UK (and available for voluntary early adoption) by way of the ambulatory reference provision in the Merchant Shipping (Carriage of Dangerous Goods and Harmful Substances) (Amendment) Regulations 2024.

2. Summary of amendments to the IMSBC Code

2.1. Amendment 07-23 to the IMSBC Code makes several changes to the Code. The following is a summary of some of the key changes.

2.1.1. **Direct Reduced Iron (DRI)** – Following many years of discussions at the IMO including input from industry groups, a new cargo schedule has

been added alongside the three existing ones for bulk DRI cargoes (see 2.2 below).

2.1.2. Electric arc furnace dust – As scrap-based steel production using electric arc furnaces (EAF) has increased around the world, there has been a corresponding increase in shipments of the dust recovered by the exhaust gas treatment plants of such steelmaking furnaces. Accordingly, a new cargo schedule for “ELECTRIC ARC FURNACE DUST, PELLETIZED” has been introduced (see 2.3 below).

2.1.3. Fish Meal - The existing individual schedule for “FISH MEAL (FISH SCRAP), STABILIZED UN 2216 Anti-oxidant treated” has been modified to align the stabilization requirements for fish meal in the IMSBC Code with the respective requirements in the IMDG Code. (Special provision 308 of the IMDG Code was previously amended to reduce the amount of ethoxyquin used for the stabilization of fish meal against self-heating and to permit a replacement of ethoxyquin by tocopherol.) The revised Bulk Cargo Shipping Name is “FISH MEAL (FISH SCRAP), STABILIZED Anti-oxidant treated”.

2.1.4. Celestine Concentrate – Following submissions to the IMO from Spain on a new / unlisted bulk cargo, the Bulk Cargo Shipping Name “CELESTINE CONCENTRATE” has been added under the heading to the individual schedule for “Mineral concentrates” in appendix 1 of the IMSBC Code.

2.1.5. The following new schedules have been added to Appendix 1 of the Code:

Cargo	Group
BARYTE, FLOTATION CHEMICAL GRADE	A
BROWN FUSED ALUMINA	C
CRUSHED GRANODIORITE FINES	A
DIRECT REDUCED IRON (D) (By-product fines with moisture content of at least 2%)	A and B
DUNITE	C
DUNITE FINES	A
ELECTRIC ARC FURNACE DUST, PELLETIZED	A and B

Cargo	Group
GROUND GRANULATED BLAST FURNACE SLAG POWDER	A
MAGNESITE FINES	A
POTASSIUM NITRATE	C
SODIUM NITRATE	C
SODIUM NITRATE AND POTASSIUM NITRATE MIXTURE	C

2.2. DIRECT REDUCED IRON (D) (By-product fines with moisture content of at least 2%) – This is a newly developed cargo schedule to cover the by-product fines from the manufacturing and handling of DRI (A) hot-moulded briquettes and/or DRI (B) lumps, pellets and cold-moulded briquettes. The cargo is classed as Group A since it may liquefy if shipped at a moisture content in excess of its transportable moisture limit (TML). The cargo is also classed as Group B because it reacts with air, fresh water and seawater to produce flammable hydrogen gas and heat.

Given the well-documented risks of overheating, fire and explosion associated with the carriage of DRI fines, the new cargo schedule is lengthy and includes an appendix which covers additional precautions to be taken prior to and during loading, development of a time-based gas prediction curve, emergency measures in case of high cargo temperature and the duties of the cargo technician (n.b. it is recommended that an experienced cargo technician is appointed by the shipper to be on board the ship during loading and throughout the voyage).

Cargo temperatures and hydrogen concentration need to be closely monitored during transit and given the vital role of ventilation in controlling the concentration of hydrogen gas in cargo holds, voyage planning must be optimized to avoid situations where the mechanical ventilation system cannot be used due to rough seas.

2.3. ELECTRIC ARC FURNACE DUST, PELLETIZED – This newly listed dry bulk cargo originates from scrap-based steel production where the material is filtered out from the exhaust gas of an electric arc furnace (EAF). It has a reddish-brown colour and is pelletized to reduce dust during handling. It is also known as EAF dust, steel mill dust and red dust.

EAF dust contains 20 to 45% zinc and is used as a raw material for zinc recovery. It also contains iron (up to 20%), calcium oxide and lead (up to 3%). Since lead has well-known adverse health effects, zinc is very toxic to aquatic life and calcium oxide has an alkaline reaction in water, the

cargo is accordingly classed as Group B with various hazards and precautions detailed in the cargo schedule. Although the cargo itself has a low fire risk, in the case of exposure to fire it is noted that the cargo may release toxic metal oxide smoke/metal fumes. The cargo is also classed as Group A since it may liquefy if shipped at a moisture content in excess of its transportable moisture limit (TML).

3. Summary of amendments to the IMDG Code

3.1. Amendment 42-24 makes multiple amendments to the IMDG Code, many of which are editorial in nature, are harmonisations with the UN Model Regulations on the Transport of Dangerous Goods, or are updates to references to, for example, ISO standards. Some amendments will have a greater impact than others. Amendment 42-24 is a consolidated edition of the IMDG Code, meaning that the full text of the Code is replaced. The annex to Resolution MSC.556(108) contains the full revised text of the Code. The following is a summary of some of the key changes.

3.2. **Marine Pollutants:** the text of 2.10.2.7 of the IMDG Code is amended to explain more clearly the requirements for marine pollutants packaged in single or combination packagings containing a net quantity per single or inner packaging of 5 L or less for liquids or having a net mass per single or inner packaging of 5 kg or less for solids. Associated with this amendment, Special Provision 375 is added to the IMDG Code. The purpose of this amendment is to more clearly set out the existing requirement, not introduce any new requirements.

3.3. **Electric Vehicles:** new UN numbers are added for VEHICLE, LITHIUM ION BATTERY POWERED (UN 3556), VEHICLE, LITHIUM METAL BATTERY POWERED (UN 3557) and VEHICLE, SODIUM ION BATTERY POWERED (UN 3558), all as Class 9 and with consequential amendments to other sections of the Code.

3.4. **New UN Numbers:** other new UN numbers added for the 42-24 amendment to the IMDG Code are: UN 3551 SODIUM ION BATTERIES with organic electrolyte; UN 3552 SODIUM ION BATTERIES CONTAINED IN EQUIPMENT or SODIUM ION BATTERIES PACKED WITH EQUIPMENT, with organic electrolyte, Class 9; UN 3553 DISILANE, Class 2.1; UN 3554 GALLIUM CONTAINED IN MANUFACTURED ARTICLES, Class 8; UN 3555 TRIFLUOROMETHYLTETRAZOLE SODIUM SALT IN ACETONE, with not less than 68% acetone, by mass, Class 3; UN 3559 FIRE SUPPRESSANT DISPERSING DEVICES, Class 9; and UN 3560

TETRAMETHYLAMMONIUM HYDROXIDE AQUEOUS SOLUTION with not less than 25% tetramethylammonium hydroxide, Class 6.1.

3.5. Special Provisions: a number of new Special Provisions (SP) are added to the 42-24 amendment to the Code, along with modifications to existing SPs, in particular as consequential amendments to the addition of the new UN numbers for electric vehicles listed in 3.3 and sodium ion batteries listed in 3.4 above. For example, SP961 and SP962 are amended to reflect the addition of the new UN 3556, 3557 and 3558; new SP977 is added as a consequence of the new UN 3558; and new SP978 and SP979 are added to improve provisions in the Code for UN 1361 CARBON of animal or vegetable origin and UN 1362 CARBON, ACTIVATED.

3.6. Devices containing dangerous goods, which are in use or intended for use during transport: the text of 5.5.4 of the IMDG Code is amended to provide more detail on requirements for devices such as data loggers, sensors and cargo tracking devices which are attached to or placed in packages, overpacks, bulk containers, freight containers or other types of cargo transport units. The amended text details what provisions of the Code apply to such devices and what standards such devices should meet.

3.7. Stowage plan: the text of 5.4.3.1 of the IMDG Code (Documentation required aboard the ship) is amended to explain that, where a stowage plan is used in place of a special list or manifest for the carriage of dangerous goods or marine pollutants, that stowage plan must identify by primary hazard class(es) and subsidiary hazard(s) and set out the location of all dangerous goods and marine pollutants. The purpose of this amendment is to set out more clearly the existing requirement, not introduce any new requirements.

3.8. Carbon: for CARBON, animal or vegetable origin UN 1361, SP925 and SP223 are deleted from column 6 of the Dangerous Goods List and the new SP978 is added for both packing groups. SP978 sets out a number of requirements, which include: that the UN N.4 test for self-heating cannot be used to exempt UN 1361 from the Code; a minimum weathering period for the unpacked cargo that applies before it can be packaged; a minimum head room requirement when packed in a CTU and a maximum stowage height of the packages (1.5m); and a maximum temperature of the product on the day of packing. For CARBON, ACTIVATED UN 1362, SP925 and SP223 are deleted from column 6 of the Dangerous Goods List and the new SP979 is added. SP979 sets out the circumstances in which the provisions of the Code (except 5.4.4.2) do not apply to steam activated and chemically activated carbon.

3.9. Certificates exempting a substance, material or article: A new paragraph 5.4.4.2 is added to the Code (Other required information and documentation). This clarifies that where a Special Provision refers to a certificate which exempts a substance, material or article from the Code, that certificate shall be submitted together with the cargo information required by SOLAS regulation VI/2.

3.10. The 2024 edition of the IMDG Code incorporating amendment 42-24 will include in the margin a symbol which identifies sections of the Code which are new, which have been amended, or where text has been deleted.

4. Application Dates

IMSBC Code

4.1. The 07-23 amendment to the IMSBC Code will enter into force on an international basis on 1 January 2025, having already been available for voluntary use since 1 January 2024.

4.2. From 1 January 2025, solid bulk cargoes shall be shipped only in accordance with the IMSBC Code incorporating the 07-23 amendment.

4.3. The IMSBC Code is available from the IMO, 1 Albert Embankment, London SE1 7SR. The Annex to IMO Resolution MSC.539(107) contains a consolidated text including amendments (07-23) to the Code.

IMDG Code

4.4. From 1 January 2025 until 31 December 2025, dangerous goods and marine pollutants in packaged form must be shipped either:

a) In accordance with the mandatory IMDG Code 2022 Edition incorporating Amendment 41-22; or

b) In accordance with the IMDG Code 2024 Edition incorporating Amendment 42-24.

4.5. On or after 1 January 2026, dangerous goods and marine pollutants in packaged form must be shipped only in accordance with the IMDG Code 2024 Edition incorporating Amendment 42-24.

More information

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